# Canon Bruce

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## EDUCATION

**Vanderbilt University** Expected Graduation: May 2028
**Nashville, TN**
B.E. Electrical and Computer Engineering, Minor in Computer Science
Dean's List Spring 2026 | GPA: 3.34/4.00
**Relevant Coursework:** Electronics, Analog Circuits, Intermediate Software Design, Linear Algebra, Differential Equations

## PROFESSIONAL EXPERIENCE

**Sitevue AI** **Franklin, TN**
*Electrical Engineer Intern* *June 2026 - August 2026*
- Developed a Linux-based robotics platform integrating computer vision, AI decision-making, and precision arm control to automate tile identification and painting across large-scale industrial environments.
- Travelled to industry trade shows to debut the robotic arm, delivering live technical demonstrations to prospective clients.
- Collaborated with engineering leadership to develop autonomous robotics solutions and support deployment opportunities.

**Legacy Audio** **Springfield, IL**
*Software Engineer Intern* *March 2026 - May 2026*
- Architected a Raspberry Pi-based Qt touchscreen interface with 10+ user controls, replacing physical controls on DSP unit.
- Led end-to-end touchscreen integration efforts, optimizing display configuration, hardware interfacing, and embedded software deployment to deliver a fully functional prototype for live product demonstration.
- Contributed to the transition from outsourced to in-house DSP production, reducing per-unit costs from $2,500 to $1,200.

**Vision Care Associates** **Springfield, IL**
*Lab Technician Intern* *May 2024 - August 2025*
- Operated precision optical machinery to fulfill 500+ patient lens prescriptions, maintaining throughput with zero quality escapes.
- Streamlined daily lab workflows by proactively communicating status updates to senior technicians, reducing delays by 20%.
- Cross-trained across multiple stages of the lens fabrication process, increasing operational flexibility within the lab.

## PROJECTS & EXTRACURRICULAR ACTIVITIES

**Quantitative Portfolio Risk & Stress-Testing Engine** **Nashville, TN**
*Python, NumPy, pandas, SciPy, Streamlit* *June 2026 – Present*
- Engineered a Monte Carlo risk engine simulating 100,000 correlated multi-asset portfolio scenarios using Cholesky decomposition and historical covariance matrices to model market uncertainty and portfolio performance
- Developed a quantitative stress-testing framework incorporating market crashes, volatility shocks, and asset correlation breakdowns; calculated Value at Risk (VaR), Conditional VaR, and maximum drawdown to quantify downside exposure.

**SiliconIQ - Semiconductor Product Strategy & Investment Analytics** **Nashville, TN**
*Python, SQL, pandas, Streamlit, Excel* *August 2026 - Present*
- Analyzed NVIDIA and AMD financials, GPU benchmarks, and R&D spending to evaluate AI accelerator market positioning.
- Developed a Python investment simulator modeling GPU pricing, market demand, and ROI across multiple growth scenarios.
- Built an interactive strategy dashboard integrating financial forecasts, product performance, and investment recommendations.

**Theta Tau, Professional Engineering Fraternity, Vanderbilt University** **Nashville, TN**
*Intramural Chair* *August 2026 – Present*
- Coordinate intramural programming across 6 teams and 5 sports for 100+ member professional engineering fraternity.
- Introduced a points-based reward system tying event attendance and team performance to a semester-end prize, incentivizing consistent participation and driving stronger team spirit and engagement across intramural programming.
- Grew chapter intramural participation from 30% to 50% by restructuring team rosters and streamlining scheduling processes.

**ECE 2123, Digital Systems, Vanderbilt University** **Nashville, TN**
*Teaching Assistant* *August 2026 - Present*
- Lead weekly lab sections for over 40 students, guiding hands-on hardware debugging and SystemVerilog implementation.
- Run group study sessions each week, helping 20+ students work through digital logic design problems and prepare for exams.

## SKILLS & INTERESTS

**Programming & Data Analytics:** Python, SQL, C++, Java, JavaScript, pandas, NumPy, Excel, MATLAB
**Engineering Tools & Platforms:** Verilog, Quartus, SPICE, CAD, Claude Code
**Interests:** Fitness & human performance, NBA analytics, caffeinated beverage connoisseur, Pokémon, SEC basketball & football